TAMIL NADU STATE BOARD - STANDARD 9 MATHEMATICS

CHAPTER 2: REAL NUMBERS

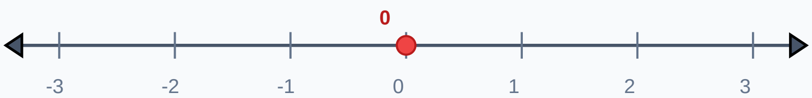
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Mathematics
Chapter Name:	Real Numbers

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

UNIT ROADMAP: REAL NUMBERS LINE CLASSIFICATION



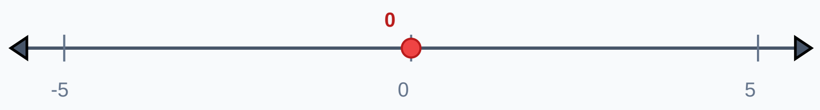
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Rational Number	A number that can be expressed in the form p/q, where p and q are integers and $q \neq 0$. Its decimal form is terminating or recurring.
Irrational Number	A number that cannot be written in the form p/q. Its decimal expansion is non-terminating and non-recurring, like π or $\sqrt{2}$.
Real Numbers	The collection of all rational and irrational numbers combined, denoted by R.
Surd	An irrational root of a rational number. For example, $\sqrt{a}$ is a surd of order 2 if a is rational but not a perfect square.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

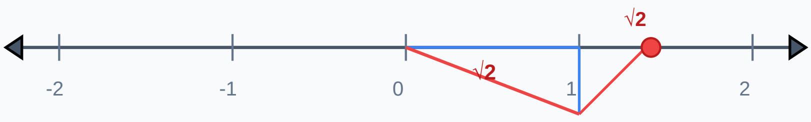
GLOSSARY: INTEGER SPACING REPRESENTATION



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: LOCATING IRRATIONAL ROOT 2 ON LINE



Essential Formulas, Laws & Identities:

- Laws of Radicals: (a^n)^(1/n) = a
- Radical multiplication: a^(1/n) * b^(1/n) = (ab)^(1/n)
- Conjugate of (a + √b) is (a - √b). Their product is a^2 - b.
- Scientific Notation: a * 10^n where 1 ≤ a < 10 and n is an integer.

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Express 0.333... in p/q form.

Step-by-step Solution:

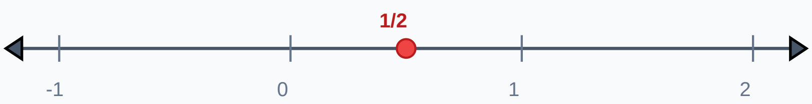
Let $x = 0.333...$ (Eq 1) $10x = 3.333...$ (Eq 2) Subtract Eq 1 from Eq 2: $9x = 3 \Rightarrow x = 3/9 = 1/3$.

Example 2: Rationalise the denominator of $1 / (3 + \sqrt{2})$.

Step-by-step Solution:

Multiply numerator and denominator by conjugate $(3 - \sqrt{2})$: $1*(3 - \sqrt{2}) / [(3 + \sqrt{2})(3 - \sqrt{2})] = (3 - \sqrt{2}) / (9 - 2) = (3 - \sqrt{2}) / 7$.

EXAMPLE: LOCATING FRACTIONS ON LINE



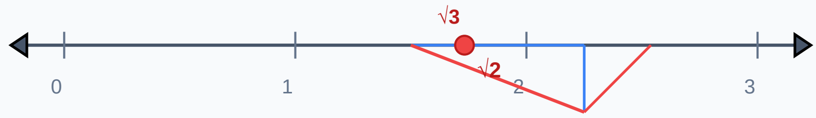
HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the rationalising factor of 5^(1/3).

Analytical Solution Strategy:
We multiply by 5^(2/3) to make the exponent 1: 5^(1/3) * 5^(2/3) = 5^(1/3 + 2/3) = 5^1 = 5. So the rationalising factor is 5^(2/3) or ■25.

HOTS: LOCATING SURDS ON LINE SCALE



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Represent $\sqrt{2}$ on the number line.

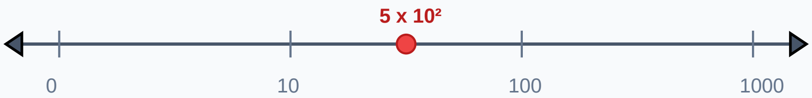
Q2 (2-Mark): Simplify: $\sqrt{12} + \sqrt{27} - \sqrt{48}$.

Q3 (2-Mark): Express 0.0000456 in scientific notation.

Q4 (5-Mark): Rationalise: $2 / (\sqrt{5} - \sqrt{3})$.

Q5 (5-Mark): Identify whether $\sqrt{8}$ is rational or irrational.

PRACTICE: SCIENTIFIC SCALES RANGE



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Acoustic Engineering

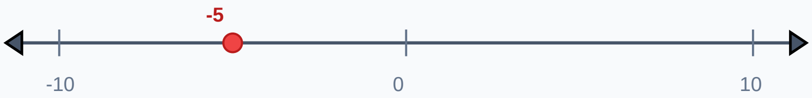
Equal temperament musical scales use frequencies scaled by $2^{(1/12)}$ (an irrational number) to tune instruments.

Application Case: Physics Scales

Scientific notation allows physicists to handle massive scales, like the mass of the Earth (5.972×10^{24} kg) without calculation errors.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: THERMOMETER SCALING SYSTEMS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

REVISION: SUMMARY NUMBER LINE SCALE